PROJECT

COUNTRIES

```
-- Basic Exploration
------------------------------------------------------------
-- Preview first 10 rows
SELECT *
FROM SH.COUNTRIES
FETCH FIRST 10 ROWS ONLY;
```

Output:

	COUNTRY_ID	COUNTRY_ISO_COD	COUNTRY_NAME	COUNTRY_SUBREGI	COUNTRY_SUBREGI	COUNTRY_REGION	COUNTRY_REGION_	COUNTRY_TOTAL	COUNTRY_TOTAL_I
1	52790	US	United States of Ame	Northern America	52797	Americas	52801	World total	52806
2	52776	DE	Germany	Western Europe	52799	Europe	52803	World total	52806
3	52789	GB	United Kingdom	Western Europe	52799	Europe	52803	World total	52806
4	52784	NL	Netherlands	Western Europe	52799	Europe	52803	World total	52806
5	52780	IE	Ireland	Western Europe	52799	Europe	52803	World total	52806
6	52777	DK	Denmark	Western Europe	52799	Europe	52803	World total	52806
7	52779	FR	France	Western Europe	52799	Europe	52803	World total	52806
8	52778	ES	Spain	Western Europe	52799	Europe	52803	World total	52806
9	52788	TR	Turkey	Western Europe	52799	Europe	52803	World total	52806
10	52786	PL	Poland	Eastern Europe	52795	Europe	52803	World total	52806

```
-- Count total countries
SELECT COUNT(*) AS total_countries
FROM SH.COUNTRIES;
```

Output:

	TOTAL_COUNTRIES
1	35

```
-- Count distinct country IDs
SELECT COUNT(DISTINCT COUNTRY_ID) AS distinct_country_ids
FROM SH.COUNTRIES;
```

Output:

	DISTINCT_COUNTRY_IDS
1	35

```
------------------------------------------------------------
-- NULL Value Checks
------------------------------------------------------------
```

```sql
SELECT
    SUM(CASE WHEN COUNTRY_NAME IS NULL THEN 1 ELSE 0 END) AS null_country_name,
    SUM(CASE WHEN COUNTRY_REGION IS NULL THEN 1 ELSE 0 END) AS null_country_region,
    SUM(CASE WHEN COUNTRY_ISO_CODE IS NULL THEN 1 ELSE 0 END) AS null_iso_code
FROM SH.COUNTRIES;
```

Output:

	NULL_COUNTRY_NAME	NULL_COUNTRY_REGION	NULL_ISO_CODE
1	0	0	0

```sql
-- Data Cleaning If Data as Null Values (EXAMPLE)
------------------------------------------------------------
-- Replace missing country name with 'Unknown'
UPDATE SH.COUNTRIES
SET COUNTRY_NAME = 'Unknown'
WHERE COUNTRY_NAME IS NULL;

-- Replace missing region with 'Unclassified'
UPDATE SH.COUNTRIES
SET COUNTRY_REGION = 'Unclassified'
WHERE COUNTRY_REGION IS NULL;

------------------------------------------------------------
-- Insights
------------------------------------------------------------
-- Number of countries per region
SELECT COUNTRY_REGION, COUNT(*) AS country_count
FROM SH.COUNTRIES
GROUP BY COUNTRY_REGION
ORDER BY country_count DESC;
```

Output:

	COUNTRY_REGION	COUNTRY_COUNT
1	Europe	14
2	Americas	6
3	Asia	6
4	Middle East	4
5	Africa	3
6	Oceania	2

```sql
-- Countries grouped by subregion
SELECT COUNTRY_SUBREGION, COUNT(*) AS country_count
FROM SH.COUNTRIES
GROUP BY COUNTRY_SUBREGION
ORDER BY country_count DESC;
```

Output:

	COUNTRY_SUBREGION	COUNTRY_COUNT
1	Western Europe	11
2	Asia	6
3	Middle East	4
4	Eastern Europe	3
5	Northern America	3
6	Southern America	3
7	Africa	3
8	Australia	2

```sql
------------------------------------------------------------
-- Relationships with Fact Tables
------------------------------------------------------------
-- Sales by country region
SELECT co.COUNTRY_REGION, SUM(s.AMOUNT_SOLD) AS total_sales
FROM SH.SALES s
JOIN SH.CUSTOMERS cu ON s.CUST_ID = cu.CUST_ID
JOIN SH.COUNTRIES co ON cu.COUNTRY_ID = co.COUNTRY_ID
GROUP BY co.COUNTRY_REGION
ORDER BY total_sales DESC;
```

Output:

	COUNTRY_REGION	TOTAL_SALES
1	Americas	55647981.76
2	Europe	28319579.56
3	Asia	10274801.98
4	Oceania	3962564.08
5	Middle East	903.83

```sql
-- Promotions impact by country
SELECT co.COUNTRY_NAME, COUNT(DISTINCT s.PROMO_ID) AS promo_count
```

```
FROM SH.SALES s
JOIN SH.CUSTOMERS cu ON s.CUST_ID = cu.CUST_ID
JOIN SH.COUNTRIES co ON cu.COUNTRY_ID = co.COUNTRY_ID
GROUP BY co.COUNTRY_NAME
ORDER BY promo_count DESC;
```

Output:

	COUNTRY_NAME	PROMO_COUNT
1	United States of Ame	4
2	Argentina	4
3	Japan	4
4	United Kingdom	4
5	Denmark	4
6	Singapore	4
7	Canada	4
8	Italy	4
9	Germany	4
10	Australia	4
11	France	4

	COUNTRY_NAME	PROMO_COUNT
8	Italy	4
9	Germany	4
10	Australia	4
11	France	4
12	Brazil	3
13	China	3
14	Poland	3
15	Spain	3
16	Turkey	2
17	Saudi Arabia	2
18	New Zealand	1

CHANNELS

```
-- Basic Exploration
------------------------------------------------------------
-- Preview first 10 rows
```

```sql
SELECT *
FROM SH.CHANNELS
FETCH FIRST 10 ROWS ONLY;
```

Output:

	CHANNEL_ID	CHANNEL_DESC	CHANNEL_CLASS	CHANNEL_CLASS_ID	CHANNEL_TOTAL	CHANNEL_TOTAL_ID
1	3	Direct Sales	Direct	12	Channel total	1
2	9	Tele Sales	Direct	12	Channel total	1
3	5	Catalog	Indirect	13	Channel total	1
4	4	Internet	Indirect	13	Channel total	1
5	2	Partners	Others	14	Channel total	1

```sql
-- Count total channels
SELECT COUNT(*) AS total_channels
FROM SH.CHANNELS;
```

Output:

	TOTAL_CHANNELS
1	5

```sql
-- Count distinct channel IDs
SELECT COUNT(DISTINCT CHANNEL_ID) AS distinct_channel_ids
FROM SH.CHANNELS;
```

Output:

	DISTINCT_CHANNEL_IDS
1	5

```sql
------------------------------------------------------------
-- Profiling
------------------------------------------------------------
-- Channels grouped by class
SELECT CHANNEL_CLASS, COUNT(*) AS channel_count
FROM SH.CHANNELS
GROUP BY CHANNEL_CLASS
ORDER BY channel_count DESC;
```

Output:

	CHANNEL_CLASS	CHANNEL_COUNT
1	Direct	2
2	Indirect	2
3	Others	1

```
-- Channels grouped by description
SELECT CHANNEL_DESC, COUNT(*) AS channel_count
FROM SH.CHANNELS
GROUP BY CHANNEL_DESC
ORDER BY channel_count DESC;
```

Output:

	CHANNEL_DESC	CHANNEL_COUNT
1	Direct Sales	1
2	Tele Sales	1
3	Partners	1
4	Internet	1
5	Catalog	1

```
------------------------------------------------------------
-- NULL Value Checks
------------------------------------------------------------
SELECT
    SUM(CASE WHEN CHANNEL_DESC IS NULL THEN 1 ELSE 0 END) AS null_channel_desc,
    SUM(CASE WHEN CHANNEL_CLASS IS NULL THEN 1 ELSE 0 END) AS null_channel_class
FROM SH.CHANNELS;
```

Output:

	NULL_CHANNEL_DESC	NULL_CHANNEL_CLASS
1	0	0

```
-- Data Cleaning(IF NULL VALUES)
------------------------------------------------------------
-- Replace missing channel description with 'Unknown'
UPDATE SH.CHANNELS
SET CHANNEL_DESC = 'Unknown'
WHERE CHANNEL_DESC IS NULL;
```

```sql
-- Replace missing channel class with 'Unclassified'
UPDATE SH.CHANNELS
SET CHANNEL_CLASS = 'Unclassified'
WHERE CHANNEL_CLASS IS NULL;

------------------------------------------------------------
-- Insights
------------------------------------------------------------
-- Sales by channel
SELECT ch.CHANNEL_DESC, SUM(s.AMOUNT_SOLD) AS total_sales
FROM SH.SALES s
JOIN SH.CHANNELS ch ON s.CHANNEL_ID = ch.CHANNEL_ID
GROUP BY ch.CHANNEL_DESC
ORDER BY total_sales DESC;
```

Output:

	CHANNEL_DESC	TOTAL_SALES
1	Direct Sales	57875260.6
2	Partners	26346342.32
3	Internet	13706802.03
4	Tele Sales	277426.26

```sql
-- Average unit price by channel
SELECT ch.CHANNEL_DESC, AVG(co.UNIT_PRICE) AS avg_unit_price
FROM SH.COSTS co
JOIN SH.CHANNELS ch ON co.CHANNEL_ID = ch.CHANNEL_ID
GROUP BY ch.CHANNEL_DESC
ORDER BY avg_unit_price DESC;
```

Output:

	CHANNEL_DESC	AVG_UNIT_PRICE
1	Internet	157.75791513962733
2	Direct Sales	145.60868248632636
3	Partners	144.92727266148

```sql
-- Channels linked to promotions
SELECT ch.CHANNEL_DESC, COUNT(DISTINCT s.PROMO_ID) AS promo_count
FROM SH.SALES s
JOIN SH.CHANNELS ch ON s.CHANNEL_ID = ch.CHANNEL_ID
```

```
GROUP BY ch.CHANNEL_DESC
ORDER BY promo_count DESC;
```

Output:

	CHANNEL_DESC	PROMO_COUNT
1	Direct Sales	3
2	Internet	3
3	Partners	2
4	Tele Sales	1

PROMOTIONS

```
-------------------------------------------------------------
-- Basic Exploration
-------------------------------------------------------------
-- Preview first 10 rows
SELECT *
FROM SH.PROMOTIONS
FETCH FIRST 10 ROWS ONLY;
```

Output:

	PROMO_ID	PROMO_NAME	PROMO_SUBCATEG	PROMO_SUBCATEG	PROMO_CATEGORY	PROMO_CATEGORY	PROMO_COST	PROMO_BEGIN_DAT	PROMO_END_DATE	PROMO_TOTAL	PI
1	375	TV promotion #13-3	TV program sponsor	13	TV	3	96500	12/23/2021, 12:00:	1/23/2022, 12:00:0	Promotion total	
2	380	TV promotion #13-3	TV program sponsor	13	TV	3	40700	3/16/2020, 12:00:0	6/16/2020, 12:00:0	Promotion total	
3	387	TV promotion #13-3	TV program sponsor	13	TV	3	7100	10/24/2020, 12:00:	1/24/2021, 12:00:0	Promotion total	
4	399	TV promotion #13-3	TV program sponsor	13	TV	3	36700	6/13/2019, 12:00:0	8/13/2019, 12:00:0	Promotion total	
5	414	TV promotion #13-4	TV program sponsor	13	TV	3	44700	10/5/2020, 12:00:0	1/5/2021, 12:00:00	Promotion total	
6	415	TV promotion #13-4	TV program sponsor	13	TV	3	60600	12/6/2021, 12:00:0	1/6/2022, 12:00:00	Promotion total	
7	426	TV promotion #13-4	TV program sponsor	13	TV	3	7700	6/2/2019, 12:00:00	8/2/2019, 12:00:00	Promotion total	
8	427	TV promotion #13-4	TV program sponsor	13	TV	3	29400	10/9/2020, 12:00:0	1/9/2021, 12:00:00	Promotion total	
9	439	TV promotion #13-4	TV program sponsor	13	TV	3	39700	6/11/2019, 12:00:0	8/11/2019, 12:00:0	Promotion total	
10	440	TV promotion #13-4	TV program sponsor	13	TV	3	78200	10/16/2020, 12:00:	1/16/2021, 12:00:0	Promotion total	

```
-- Count total promotions
SELECT COUNT(*) AS total_promotions
FROM SH.PROMOTIONS;
```

Output:

	TOTAL_PROMOTIONS
1	503

```
-- Count distinct promotion IDs
SELECT COUNT(DISTINCT PROMO_ID) AS distinct_promo_ids
```

```
FROM SH.PROMOTIONS;
```

Output:

	DISTINCT_PROMO_IDS
1	503

```
-- Profiling
-------------------------------------------------------------
-- Promotions grouped by category
SELECT PROMO_CATEGORY, COUNT(*) AS promo_count
FROM SH.PROMOTIONS
GROUP BY PROMO_CATEGORY
ORDER BY promo_count DESC;
```

Output:

	PROMO_CATEGORY	PROMO_COUNT
1	TV	115
2	newspaper	110
3	internet	85
4	magazine	65
5	post	65
6	flyer	40
7	radio	20
8	ad news	2
9	NO PROMOTION	1

```
-- Promotions grouped by subcategory
SELECT PROMO_SUBCATEGORY, COUNT(*) AS promo_count
FROM SH.PROMOTIONS
GROUP BY PROMO_SUBCATEGORY
ORDER BY promo_count DESC;
```

Output:

	PROMO_SUBCATEG	PROMO_COUNT
1	TV program sponsor:	75
2	coupon news	65
3	online discount	50
4	downtown billboard	40
5	coupon magazine	35
6	ad news	30
7	hospital flyer	25
8	TV commercial	25
9	freeway billboard	20
10	manufacture rebate r	20
11	ad	20

	PROMO_SUBCATEG	PROMO_COUNT
12	loyal customer discou	15
13	radio program spons	15
14	promotion movie	15
15	manufacture rebate r	15
16	general flyer	10
17	ad magazine	10
18	billboard	5
19	household flyer	5
20	radio commercial	5
21	newspaper	2
22	NO PROMOTION	1

```
-------------------------------------------------------------
-- NULL Value Checks
-------------------------------------------------------------
SELECT
    SUM(CASE WHEN PROMO_NAME IS NULL THEN 1 ELSE 0 END) AS null_promo_name,
    SUM(CASE WHEN PROMO_CATEGORY IS NULL THEN 1 ELSE 0 END) AS null_promo_category,
    SUM(CASE WHEN PROMO_COST IS NULL THEN 1 ELSE 0 END) AS null_promo_cost
FROM SH.PROMOTIONS;
```

Output:

	NULL_PROMO_NAME	NULL_PROMO_CATEGORY	NULL_PROMO_COST
1	0	0	0

```
-- Data Cleaning (examples)
------------------------------------------------------------
-- Replace missing promotion name with 'Unknown'
UPDATE SH.PROMOTIONS
SET PROMO_NAME = 'Unknown'
WHERE PROMO_NAME IS NULL;

-- Replace missing category with 'Unclassified'
UPDATE SH.PROMOTIONS
SET PROMO_CATEGORY = 'Unclassified'
WHERE PROMO_CATEGORY IS NULL;

------------------------------------------------------------
-- Insights
------------------------------------------------------------
-- Total promotion spend by category
SELECT PROMO_CATEGORY, SUM(PROMO_COST) AS total_spend
FROM SH.PROMOTIONS
GROUP BY PROMO_CATEGORY
ORDER BY total_spend DESC;
```

Output:

	PROMO_CATEGORY	TOTAL_SPEND
1	newspaper	5680500
2	TV	5320000
3	internet	4256200
4	magazine	3402700
5	post	3313100
6	flyer	2163400
7	radio	1095000
8	NO PROMOTION	0
9	ad news	0

```
-- Top 10 most expensive promotions
SELECT PROMO_ID, PROMO_NAME, PROMO_COST
FROM SH.PROMOTIONS
ORDER BY PROMO_COST DESC
```

```
FETCH FIRST 10 ROWS ONLY;
```

Output:

	PROMO_ID	PROMO_NAME	PROMO_COST
1	40	newspaper promotio	100000
2	481	TV promotion #12-4	99900
3	157	post promotion #17-	99700
4	530	TV promotion #12-5	99600
5	274	post promotion #20-	99300
6	343	post promotion #20-	99200
7	281	post promotion #20-	99100
8	122	radio promotion #32	99000
9	133	magazine promotion	98700
10	398	newspaper promotio	98500

```
-- Average promotion cost by subcategory
SELECT PROMO_SUBCATEGORY, AVG(PROMO_COST) AS avg_cost
FROM SH.PROMOTIONS
GROUP BY PROMO_SUBCATEGORY
ORDER BY avg_cost DESC;
```

Output:

	PROMO_SUBCATEGORY	AVG_COST
1	ad magazine	68980
2	household flyer	66660
3	loyal customer discount	61620
4	general flyer	58050
5	radio program sponsorship	56506.666666666664
6	coupon news	54935.38461538462
7	downtown billboard	53632.5
8	ad	51025
9	coupon magazine	50922.857142857145
10	promotion movie	50753.333333333336
11	TV commercial	50664

	PROMO_SUBCATEGORY	AVG_COST
12	ad news	50243.333333333336
13	hospital flyer	49984
14	radio commercial	49480
15	billboard	48080
16	manufacture rebate magazine	46530
17	freeway billboard	46370
18	online discount	46228
19	TV program sponsorship	43894.666666666664
20	manufacture rebate news	40160
21	NO PROMOTION	0
22	newspaper	0

```
-------------------------------------------------------------
-- Relationships with Fact Tables
-------------------------------------------------------------
-- Sales linked to promotions
SELECT p.PROMO_NAME, SUM(s.AMOUNT_SOLD) AS total_sales
FROM SH.SALES s
JOIN SH.PROMOTIONS p ON s.PROMO_ID = p.PROMO_ID
GROUP BY p.PROMO_NAME
ORDER BY total_sales DESC;
```

Output:

	PROMO_NAME	TOTAL_SALES
1	NO PROMOTION #	94504520.79
2	internet promotion #	2199380.9
3	TV promotion #13-3	1224503.26
4	post promotion #20-	277426.26

```
-- Costs linked to promotions
SELECT p.PROMO_NAME, AVG(c.UNIT_COST) AS avg_cost, AVG(c.UNIT_PRICE) AS avg_price
FROM SH.COSTS c
JOIN SH.PROMOTIONS p ON c.PROMO_ID = p.PROMO_ID
GROUP BY p.PROMO_NAME
ORDER BY avg_price DESC;
```

Output:

	PROMO_NAME	AVG_COST	AVG_PRICE
1	internet promotion #	161.42796013289038	208.14699003322258
2	TV promotion #13-3	118.36296975252063	153.07467919340056
3	NO PROMOTION #	116.90379254064393	146.96189085113164